ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (CSE) CLOUD COMPUTING

Semester: II

SUBJECT CODE: BCSI201

DESIGN THINKING

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Define the following terms: (1 × 5 = 5 Marks)

(i) What is Design Thinking? Give an example from a tech-based product.

(ii) How is Ideation used in redesigning college campus navigation apps?

(iii) How can Usability and Creativity conflicts when designing smart kitchen devices?

(iv) Why is 'Learning by Doing' essential in engineering problem-solving?

(v) What role does the 'Test Cycle' play in refining a digital product?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) Mention any three core creativity skills used in developing social media features.

(ii) How does Qualitative Research help improve wearable health tech?

(iii) Explain the role of Empathy in designing accessible metro stations.

(iv) What is Reframing? How can it help in redesigning online education systems?

(v) Explain Innovation with an example from the electric vehicle industry.

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3(i)(a) A startup wants to analyze customer complaints to redesign their delivery app. How can Grounded Theory help?

OR

(b) A design sprint is conducted to create eco-friendly packaging. Explain any three Idea Generation Approaches used during the sprint.

(ii)(a) A team is working on a wearable for mental wellness. Why is Idea Generation important? Mention two techniques that could be applied.

OR

(b) An ed-tech teacher wants students to innovate. How does linking Teaching with Learning promote Creativity?

Q.4(i)(a) A company is testing a smart shoe prototype. Explain the role of Prototyping and how it differs from Testing.

OR

(b) A public service app is being developed for differently-abled users. How can Human-Centered Design guide the process? Mention its goals.

(ii)(a) A logistics firm wants to improve delivery tracking efficiency. How does Process Knowledge support design?

OR

(b) UPI was widely adopted across India. Use the four elements of Diffusion of Innovation to explain how and why this happened. Include a labeled diagram.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5(a) A hospital is creating a health report system for patients with limited tech access. What Research Design should they follow? Explain all four types.

OR

(b) Differentiate between Creativity and Innovation using examples from Tesla and IKEA. How does Creativity impact teaching in the digital era?

Q.6(a) A mobile banking app for rural users is under development. How would you apply Observe, Describe, and Participate to understand user challenges?

OR

(b) Redesigning a railway ticket booking system using Design Thinking made it easier for all age groups. Explain how Design Thinking principles helped in this transformation.